THEORY LABORATORY FOR MODERN ROBOTICS

Solutions for Training Set 3: Manifolds

Online discussion of model solutions (sets 3 and 4) on Monday, 8 June 2020, 14:00–16:00

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1. **Understanding the concepts (SOLUTIONS ONLY GIVEN IN THE TUTORIAL)**

Mark the correct statements by ■□ and the incorrect ones by □■. You learn from your mistakes.

□■ A topology may contain sets that are neither open nor closed.

■□ Choosing all subsets of a set M defines a particular topology.

■□ A set with two elements can be equipped with exactly four different topologies.

■□ Any map $\varphi : M \to N$ is continuous if the topology on the domain M contains all subsets of M.

■□ Any map $\varphi : M \to N$ is continuous if the topology on the target space N is the set $\{\varnothing, N\}$.

□■ The only consistent choice for a topology on the real numbers is the standard topology.

□■ An arbitrary set M can be equipped with at least two more topologies than the standard topology.

■□ There always exist enough charts to completely cover a topological manifold.

■□ The chart transition map $y \circ x^{-1}$ between two charts (U, x) and (V, y) is continuous.

■□ The continuity of a curve in a manifold can be verified with or without the use of charts.

□■ Any smooth manifold is also a topological manifold, while the converse holds only for some cases.

■□ The atlas of a smooth manifold cannot contain all charts of the underlying topological manifold.

2. **Mastering the formalism**

Mastership is in the basics. Work hard to find the required short proofs. This exercise will pay off.

(a) Let $(M, \mathcal{O})$ be a topological space and S be some subset of M. Prove that the so-called *subset topology*

$$\mathcal{O}|_S := \{U \cap S \,|\, U \in \mathcal{O}\}$$

indeed defines a topology on the subset S.

<u>Solution:</u> *The set $\mathcal{O}|_S$ satisfies the first axiom for a topology for the set S since*

$$\varnothing = \varnothing \cap S \qquad \text{and} \qquad S = M \cap S$$

since $\varnothing$ on the right hand side of the first equation and M on the right hand side of the second equation both lie in $\mathcal{O}$. To show that it also satisfies the second axiom for a topology for S, assume that $A, B \in \mathcal{O}|_S$, so that – by definition of the subset topology – there are some $U, V \in \mathcal{O}$ such that $A = U \cap S$ and $B = V \cap S$. But then we have

$$A \cap B = (U \cap S) \cap (V \cap S) = (U \cap V) \cap S\,,$$

which is in $\mathcal{O}|_S$ since $U \cap V \in \mathcal{O}$ due to $\mathcal{O}$ being a topology. To show that also the third axiom for a topology for S is satisfied by the set $\mathcal{O}|_S$, assume that there is a family of sets $A_\alpha \in \mathcal{O}|_S$ for α from an arbitrary set index I, whence there must a family $U_\alpha \in \mathcal{O}$ such that $A_\alpha = U_\alpha \cap S$. But then

$$\bigcup_{\alpha \in I} A_\alpha = \bigcup_{\alpha \in I} U_\alpha \cap S = \left(\bigcup_{\alpha \in I} U_\alpha\right) \cap S\,,$$

which is in $\mathcal{O}|_S$ because $\bigcup_{\alpha \in I} U_\alpha \in \mathcal{O}$. Thus, in summary, $\mathcal{O}|_S$ qualifies as a topology for S.

(b) Let $(M, \mathcal{O}_M)$ and $(N, \mathcal{O}_N)$ be topological spaces. Define the so-called *product topology* $\mathcal{O}_{M\times N}$ on $M \times N$ as the set that contains precisely those subsets $W \subseteq M \times N$ for whose every point $w = (m, n) \in W$ there exists a $U \in \mathcal{O}_M$ with $m \in U$ and a $V \in \mathcal{O}_N$ with $n \in V$ such that $U \times V \subseteq W$. Show that $\mathcal{O}_{M\times N}$ indeed defines a topology on the cartesian product set $M \times N$. Graphically illustrate the construction for $M = N = \mathbb{R}$.

Solution: *First axiom: The empty set trivially lies in $\mathcal{O}_{M\times N}$, since any condition on its elements is automatically satisfied, because it does not have any elements. The set $W = M \times N$ lies in $\mathcal{O}_{M\times N}$, since for every point $(m, n) \in W$ we can choose $U = M \in \mathcal{O}_M$ and $V = N \in \mathcal{O}_N$ to satisfy the condition. Second axiom: Let $A, B \in \mathcal{O}_{M\times N}$. Then, by definition of the product topology, for every point $(m, n) \in A$ there exist $U_A \in \mathcal{O}_M$ and $V_A \in \mathcal{O}_N$ with $m \in U_A$, $n \in V_A$ and $U_A \times V_A \subseteq A$. Analogously there exist such sets U_B, V_B for B. But then for any point $(p, q) \in A \cap B$, one has $p \in U_A \cap U_B$, $q \in V_A \cap V_B$ and*

$$(U_A \cap U_B) \times (V_A \cap V_B) = (U_A \times V_A) \cap (U_B \times V_B) \subseteq A \cap B\,.$$

Third axiom: Consider a family of sets $A_\alpha \in \mathcal{O}_{M\times N}$ for $\alpha \in I$. Then there exist families of sets $U_\alpha \in \mathcal{O}_M$ and $V_\alpha \in \mathcal{O}_N$ such that for every point $(m, n) \in A_\alpha$ one has that $m \in U_\alpha$ and $n \in V_\alpha$ and $U_\alpha \times V_\alpha \subseteq A_\alpha$. But then for any $(p, q) \in \bigcup_{\alpha\in I} A_\alpha$ there exists an $\alpha' \in I$ such that $(p, q) \in A'_\alpha$ and thus

$$U_{\alpha'} \times V_{\alpha'} \subseteq A_{\alpha'} \subseteq \bigcup_\alpha A_\alpha\,.$$

Thus $\mathcal{O}_{M\times N}$ is a topology for the set $M \times N$.

(c) Show that the so-called *standard topology* $\mathcal{O}_s$ on $\mathbb{R}^d$ indeed satisfies the axioms of a topology.

Solution: *First axiom: The empty set $\varnothing$ automatically satisfies the conditions on any points it contains, since it contains none. The entire set $\mathbb{R}^d$ is open since $B_r(p) \subset \mathbb{R}^d$ for any $r \in \mathbb{R}^+$ and any $p \in \mathbb{R}^d$. Second axiom: Let $U, V \in \mathcal{O}_s$. Then the intersection $U \cap V$ is either empty – and thus open in the standard topology, as already shown – or it contains a point p, so that necessarily $p \in U$ and $p \in V$. But then there are $r_U \in \mathbb{R}^+$ and $r_V \in \mathbb{R}^+$ such that $B_{r_U}(p) \subseteq U$ and $B_{r_V}(p) \subseteq V$ since $U, V \in \mathcal{O}_s$. But then*

$$B_{min\{r_U, r_V\}} \subseteq U \cap V\,.$$

Third axiom: If $U_\alpha \in \mathcal{O}_s$ for α in some arbitrary index set I, then for any point p of the union $\bigcup_{\alpha\in I} U_\alpha$ there is an $\alpha' \in I$ such that $p \in U_{\alpha'}$. Now since $U_{\alpha'} \in \mathcal{O}_s$, there is an $r \in \mathbb{R}^+$ such that $B_r(p) \subseteq U_{\alpha'}$, and thus also

$$B_r(p) \subseteq \bigcup_{\alpha\in I} U_\alpha\,.$$

(d) Prove that the composition $f \circ \gamma$ of a smooth curve $\gamma : \mathbb{R} \to M$ and a smooth function $f : M \to \mathbb{R}$ on a smooth manifold $(M, \mathcal{O}, \mathcal{A})$ is again smooth.

Solution: *A function $f : M \to \mathbb{R}$ is smooth on the domain U of a chart $(U, x) \in \mathcal{A}$ if the chart representative $f \circ x^{-1} : x(U) \to \mathbb{R}$ is smooth. Since the domains of all charts in the smooth atlas $\mathcal{A}$ cover M, the smoothness of f over M is thus ensured if it is ensured for each chart in $\mathcal{A}$. Similarly, a curve $\gamma : \mathbb{R} \to M$ is smooth if its chart representative $x \circ \gamma : \mathbb{R} \to x(U)$ is smooth for every chart $(U, x) \in \mathcal{A}$. But then the composition*

$$f \circ \gamma = f \circ (x^{-1} \circ x) \circ \gamma = (f \circ x^{-1}) \circ (x \circ \gamma)$$

is smooth since the composition of the two smooth maps $x \circ \gamma : \mathbb{R} \to \mathbb{R}^d$ and $f \circ x^{-1} : \mathbb{R}^d \to \mathbb{R}$ is a smooth function $\mathbb{R} \to \mathbb{R}$.

(e) Let $(M, \mathcal{O})$ be a topological space. If nothing else is said, how would you interpret the statement that a map

$$\varphi : M \times M \to \mathbb{R}$$

is continuous?

Solution: *Continuity is defined for maps* $\varphi : N \to P$ *between any two topological manifolds* $(N, \mathcal{O}_N)$ *and* $(P, \mathcal{O}_\mathcal{P})$*. Thus one must consider* $N := M \times M$ *and* $P = \mathbb{R}$*. The question that remains is which topologies one chooses to equip* N *and* P *with. If nothing else is said, one would usually assume that one employs the topology* $\mathcal{O}$ *on* M *to equip* $N = M \times M$ *with the product topology* $\mathcal{O}_{M\times M}$ *and* $P = \mathbb{R}$ *with the standard topology on* $\mathbb{R}$ *in order to decide the continuity of the map* φ*.*

(f) Prove that the component maps $x^1, \dots, x^{\dim M}$ defined by

$$x^a : U \to \mathbb{R}\,, \qquad p \mapsto a\text{-th entry of the } \dim M\text{-tuple } x(p)$$

for any chart (U, x) in a smooth atlas $\mathcal{A}$ of a smooth manifold $(M, \mathcal{O}, \mathcal{A})$ are smooth functions on U.

Solution: *This is a cute question, because if one considers* $(M, \mathcal{O})$ *merely as a topological manifold, even the restriction that one only considers charts* (U, x) *from some smooth atlas* $\mathcal{A}$ *that is assumed to be built over* $(M, \mathcal{O})$ *does not suffice to render the question of whether* $x : U \to x(U)$ *is smooth or not meaningful in the first place. But of course, since we consider* $(M, \mathcal{O}, \mathcal{A})$ *to be a smooth manifold from the start, we can decide the smoothness of any function from* M *to* $\mathbb{R}$*. In particular, we may decide the smoothness of the individual coordinate functions* $x^a : U \to \mathbb{R}$ *derived from the chart map* x *of any chart* $(U, x) \in A$ *by checking whether the*

$$x^a \circ x^{-1} : x(U) \to \mathbb{R}$$

are smooth for all $a = 1, \dots, \dim M$*. This, however, is always the case since*

$$(x^a \circ x^{-1})(\alpha^1, \dots, \alpha^{\dim M}) = \alpha^a$$

for all $(\alpha^1, \dots, \alpha^d) \in x(U) \subseteq \mathbb{R}^{\dim M}$*, whence* $x^a \circ x^{-1}$ *is a smooth function* $x(U) \subseteq \mathbb{R}^{\dim M} \to \mathbb{R}$*. Thus, indeed, the coordinate functions* x^a *derived from the chart map* x *of any chart* (U, x) *of the smooth atlas* $\mathcal{A}$ *are all smooth functions, and only because of this it is possible to consider the differentials* $dx^a := d(x^a)$ *to build coordinate-induced bases of the cotangent spaces* T_p^*M *for any* $p \in U$*.*